MIL-G 13490 (Ord)
23 June 1954

SUPERSEDING
PA-PD-275 (Rev 1)
20 April 1954

MILITARY SPECIFICATION

GRENADE, RIFLE, HEAT, M31
METAL PARTS (LESS LUCKY ASSEMBLY)

1. SCOPE

1.1 This specification covers the metal parts (less lucky assembly) for the M31 HEAT Rifle Grenade.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, drawings, and publications, and all specifications and drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550	-General Specification for Ammunition Except Small Arms Ammunition
MIL-P-10025	-Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For

DRAWINGS

ORDNANCE CORPS

82-O-195	-Grenade, Rifle, HEAT, M31 Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M5-106	-Grenade, Rifle, HEAT, M31 Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawing and specifications.

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3.2 The metal parts (less lucky assembly) shall comply with all requirement specified on Drawing 82-0-195, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Stabilizer assembly.-Each stabilizer assembly shall withstand a load of 200 pounds without evidence of rivet failures when tested as specified in 4.5.

3.4 Hydrostatic.-Each stabilizer or stabilizer tube shall withstand a internal hydrostatic pressure of 4,500 psi for 2 seconds minimum, without evidence of cracking or deformation, when tested as specified in 4.6.

3.5 Pull.-The joint between the stabilizer tube and adapter shall withstand a minimum pull of 2,000 pounds without separating more than .02 inch, when tested as specified in 4.7.

3.6 Interior and exterior surfaces.-The inside of each body assembly shall be free of lumps of solder. The exterior surfaces of the body assembly shall be clean and free of paint except where specified on the body assembly drawing.

3.7 Cone surfaces.-The cone shall be smooth and free of scores, pits, and other surface imperfections. Slight evidence of successive draws (draw rings) on the exterior surface of the cone shall not be considered on objectionable imperfection.

3.8 Coatings.-The interior and exterior coatings specified on the drawings shall be applied in such a manner as to insure an even distribution and thorough coating.

3.9 Workmanship.-All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.-A lot shall consist of all grenade metal parts produced by one manufacturer under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, the same specification and the same specification revision.

4.2 Sampling.-Number of samples, acceptance, and rejection criteria used for determination of lot acceptance for tests as specified herein shall be in accordance with Standard Inspection Procedure ORD-SIP-M5-106.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Internal air pressure test.

4.4.1 Ogive.-Sample body assemblies shall be placed in a satisfactory fixture and the internal air pressure specified on the drawing shall be applied at open end of ogive for the period specified on the drawing. The gage used shall be capable of determining a pressure drop of .06 psi.

4.4.2 Body.-Sample body assemblies shall be placed in a satisfactory fixture and the internal air pressure specified on the drawing, shall be applied at open end of body for the period specified on the drawing. The gage used shall be capable of determining a pressure drop of .06 psi.

4.5 Test of stabilizer assembly.-Sample stabilizer assemblies shall be assembled in the fixture shown on figure 1 and tested by applying a 200-pound load along the axis of the tube assembly after riveting of the fin to the tube assembly. The load shall be applied at a rate of .100 to .125 inch per minute, inclusive. Each sample stabilizer assembly shall be observed to determine

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compliance with 3.3.

4.6 Hydrostatic test.-Each stabilizer or stabilizer tube prior to assembly to adapter shall be placed in a suitable fixture and subjected to an internal hydrostatic pressure of 4,500 psi for a period of 2 seconds, minimum. Each stabilizer tube shall be observed to determine compliance with 3.4.

4.7 Pull test.-Sample stabilizer assemblies shall be selected by the inspector from each lot and placed in a satisfactory test fixture. A load of 2,000 pounds, minimum shall be applied. Each sample stabilizer assembly shall be observed to determine compliance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, and marking.-Packaging, packing, labeling, and marking shall be as specified in Specification MIL-P-10025.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

Notice. -This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, or be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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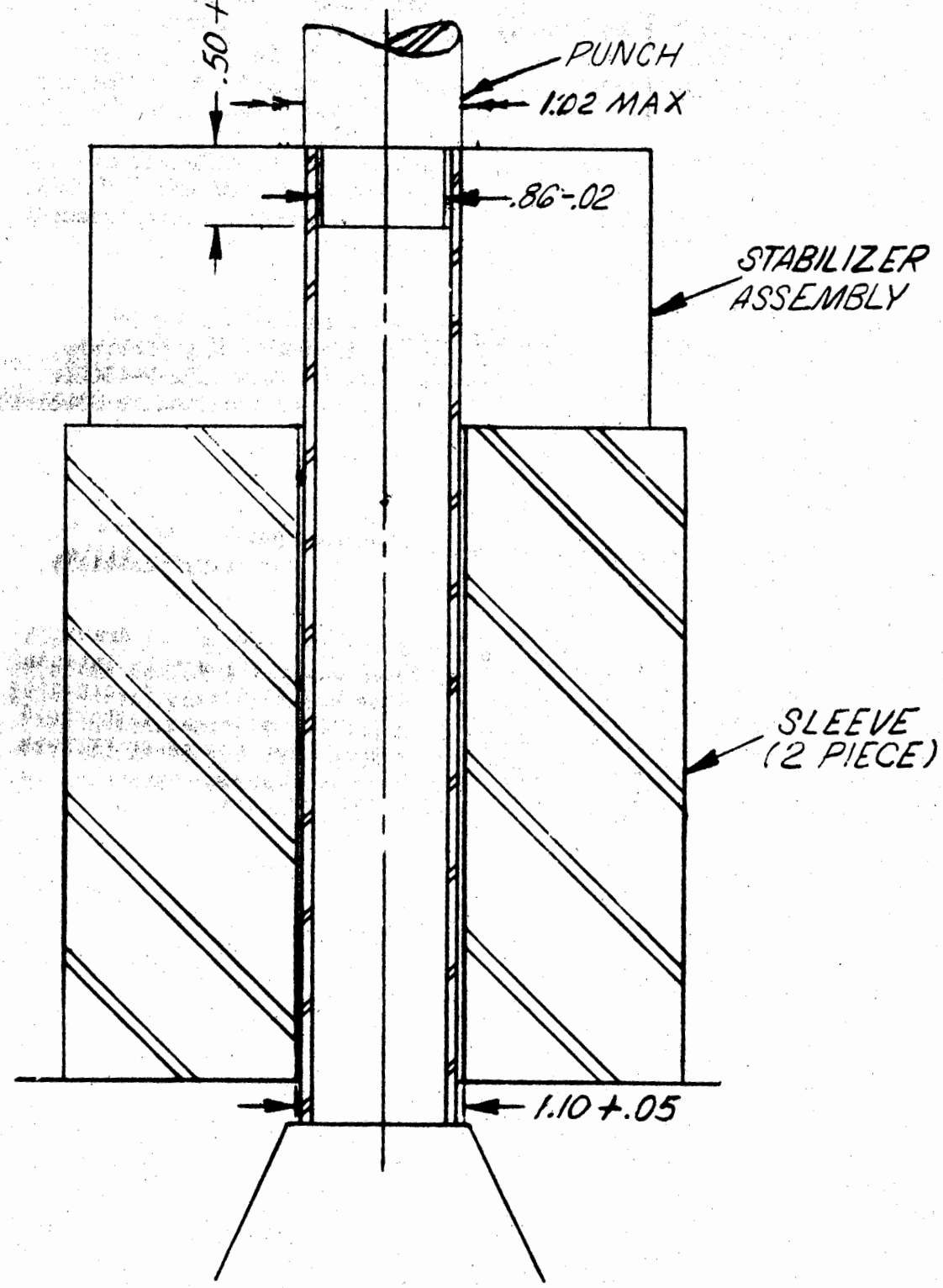


FIGURE 1

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TEST FIXTURE FOR STABILIZER ASSEMBLY